



**Fundamentally New
Immunological
Approaches In Solving
Problems of Infectious
Diseases of Viral, Bacterial
And Fungal Etiology**

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INTRODUCTION

With the global population exposed to the ongoing Coronavirus crisis, a permanent and expedient solution is needed at an affordable cost. Due to the high volume of cases, the crisis will likely manifest itself most severely in lower income and less developed communities in the coming weeks and months. Fortunately, we believe we have the lowest cost option already in hand in a drug we have long tested for treatment of AIDS and other infectious diseases.

MAIN SCIENTIFIC DISCOVERIES AND GENERATED MEDICATIONS

For the past three decades our team of researchers has proliferated in the fields of pharmaceutical, medical, and veterinary sciences. Relying on millions of clinical laboratory studies, as well as more than 5,000 in-vivo cases of human testing, we have compiled a significant statistical database, which comprises the underpinning of a deep understanding of fundamental biological mechanisms. Creativity, combined with this knowledge, has led to our breakthroughs the early diagnosis and infectious disease treatment of numerous human and animal maladies. Our extensive line of medications is based on the high dilution (HD) of active ingredients to combine maximum efficiency and efficacy.

Our primary mission drives our every interaction: "Active longevity means an assured absence of oncological diseases and efficient prophylaxis of metabolic and infectious diseases." To that end, our scientists have developed a system of measures, including both medications and unique treatment methods, focused on both the quantity and quality of life. Simply stated, our products and information help patients live both longer, and better, free of chronic disease.

Following through on our initial discoveries of more than 20 years ago, we continue to advance diagnostic tools for detection of biochemical, bio energetical and pharmacological activity of both natural and synthesized substances in HD. The result is a cutting-edge evaluation of optimum physiological outcomes that lead the field of medicine into the 21st century.

A FEW OF OUR MOST KEY SUCCESSSES

- The largest volume of research performed on farm and laboratory animals — more than five million cases;
- Breakthrough research of bioorganisms' homeostasis and redox systems;
- Development of researched-based innovative technologies leading to the production of fundamentally new drugs;
- Successful clinical studies and development of protocols for the new drugs, leading to marketplace establishment and prevention of disease and improved treatment for patients.

As we have not yet taken our portfolio commercial, see our list of pathologies in our portfolio introduction to identify the variety of areas that we have developed solutions for a broader introduction.

FUNDAMENTALLY NEW IMMUNOLOGICAL APPROACHES IN SOLVING PROBLEMS OF INFECTIOUS DISEASES OF VIRAL, BACTERIAL AND FUNGAL ETIOLOGY

THE CURRENT PARADIGM

Currently, the principles of animal and human immune protection are based on active immunization or interference against virulent strains of pathogens. These approaches are based on the development of antibodies against corpuscular or soluble components of microorganisms.

The immune protection that occurs after the corresponding disease or the introduction of vaccines is based primarily on the interaction of pathogens with phagocytes (non-specific incomplete phagocytosis). In the sequel, there is a presentation of antigens of pathogens by T-lymphocytes, activation of B-lymphocytes, which ends with the production of antibodies.

These immune defense mechanisms can be implemented with true antigens of microorganisms that have a certain molecular weight, but not with haptens that have a low molecular weight.

A full-fledged immune response in animals and humans occurs only if there is a mature immune system at a certain age in the postnatal period.

WWMA AG WAS DEVELOPED

At the same time, there is a neonatal period during which, as a rule, a full-fledged immune response cannot be carried out. The main factor of protection is non-specific phagocytosis (see also note 3).

The aim of WWMA AG's research was to develop methods for protecting the animal and human bodies from infectious agents during their initial contact.

For these purposes, endogenous and secreted components (proteins, lipids and polysaccharides) of microorganisms were used, which are haptens that are not able to activate the production of antibodies.

The introduction of the components selected by us in a certain way (using the developed means of targeted delivery to immunocompetent organs) in the neonatal period and later periods of ontogenesis provides a specific completed phagocytosis of pathogens (bacteria, fungi), as well as activation of antiviral immunity based on the components contained in the pathogens of viral diseases.

Data obtained at the first stage of research on tuberculosis, brucellosis, salmonellosis, actinomycosis, candidiasis, vector-borne gastroenteritis, swine reproductive and respiratory syndrome, as well as on influenza and several other types of coronavirus and other infectious diseases showed effective immune protection, which in many cases provides life-long resistance of animals and long-term (the period of observation for patients immunized with the flu drug is 6 years, there is reason to believe also-life-long) human resistance to infection.

Based on the results obtained, we have developed a mechanism and drugs that fix low-molecular-weight protective components of microorganisms in immune-competent organs, and methods of their application for complex protection of a person from infectious "damaging" factors.

CURRENT AREAS OF FUNDAMENTAL RESEARCH

- Study of the mechanism of maternal extrachromosomal heredity and its use in solving problems of infectious and non-communicable diseases of laboratory and agricultural (pigs, cattle) animals — as part of the work on modeling such processes in the human body with the transmission of resistance to these diseases in generations (on the mother's side).
- Simulation on laboratory and farm animals the process of forming an active T-killer immune defense system in the neonatal period and prolonging its action in the postnatal period in the phenotype.
- Development of drugs that fix in immune-competent organs low-molecular-weight protective components of microorganisms, and methods of their application for complex protection of a person from infectious diseases of bacterial, viral and fungal etiology.

As part of this work, various cell culture lines (strains) were created, which we use both for creating new drugs for preventive and therapeutic purposes, and for diagnostic technological purposes: Strain "VN-96", Strain *Aspergillus fumigatus* #6, strain *Candida albicans* # 253, Strain *Escherichia coli* B-5. These strains are patented and deposited in the state research Institute for control, standardization and certification of veterinary drugs and feed.

For the purpose of creating drugs for the treatment and prevention of coronavirus that is currently spreading, the strain "VN-96" is needed. See Note 01.

CURRENT CAPABILITIES OF THE COMPANY

1. For urgent use — a therapy-medicine, with the working name XR Coronavirus +, common antiviral and some specific anti-coronavirus part, 2-4 week therapy time, after 10 days will be clear positive result — but not certified; see Note 02.
2. Produced in a short time (2-4 weeks) — based on CC VN-96 therapy medicine with two injections as therapeutic — but does not exist in an existing laboratory animal model for the most recent coronavirus mutation.
3. Also produced in a short time — based on CC VN-96 anti-coronavirus prophylaxis medicine for vaccine, which is a single injection sufficient for immunization.

PROPHYLAXIS-THERAPEUTIC MEDICINE "XR ANTI-CORONAVIRUS+", ONE INJECTION – 5 ML AMPOULE, FINAL SOLUTION?

- Course of (temporary) prevention of healthy from Coronavirus 2019nCoV patients – 7 injections for 18 days.
- Therapeutic course — 10 injections for 21 days.
- A total of 10 ampoules of 5 ml or 50 ml of the finished product are required per course for one patient.
- For 10,000 patients, 500 liters of the finished product in 100,000 ampoules are required.
- The cost of one ampoule content is 5 euros.
- The German side presents a clean laboratory room for the production of finished products without packaging by ampoules.
- The German side provides the NaCl solution in the amount necessary for the manufacture of the agreed volume of finished products.
- WWMA AG specialists produce a ready-made product, without packaging in ampoules, at the rate of 500 liters per 10,000 patients or more, as agreed by the parties.
- The parties enter into a contract for the performance of services for the production of this drug before the departure of WWMA AG specialists.
- The German side bears all the necessary expenses for the arrival of WWMA AG specialists, their placement, pays the agreed travel expenses and provides all the necessary conditions for the production of the finished product without packaging in ampoules.
- The German side independently arranges the filling of the finished form of the drug in ampoules at the corresponding enterprise in Taiwan.
- WWMA AG specialists train specialists of the German side in the methods of using ready-made forms of drugs.

PRODUCTION OF A VACCINE

The next step after the therapeutic drug application stage is the production of a vaccine that provides long-term protection of the patient's body from the coronavirus by a single injection 2019nCoV – minimum funds will be necessary to create and produce new generation of prevention medicine, based on WWMA AG developed strain VN-96, is 12.7 mln Euro and needed 3 to 5 months maximum once WWMA AG receives funding.

STRAIN VN - 96

Strain "VN-96" – specified as transmissible gastroenteritis virus of swine — deposited in the State Research Institute of control, standardization and certification of veterinary drugs and feed, certificate of deposit of the production strain from 27.05.2009 on two names, V.Laskavyi and S.Ivanenko. The strain is patented in Russian Federation, Patent "Strain VN-96 of transmissible gastroenteritis virus of pigs used for the production of vaccines and diagnostic."

ABSTRACT 1 – ABOUT THE ESSENCE OF THE DISCOVERY AND MECHANISM OF ACTION OF XR CORONAVIRUS+

In contrast to the existing paradigm — the control mechanism of inflammation "peroxide-nitrite" ($\text{NO} + \text{H}_2\text{O}_2$), developed by WWMA AG has taken this hypothesis: the blocking mechanism of the inflammatory process (proinflammatory cytokines) through the control via the redox-system amino group (L-arginine $\rightarrow$ proline a/K) $\rightarrow$ polyamine - bioregulators long nekotoroyh RNA). Therefore, adjusting the newly created drug through the redox system, the metabolic node, L-arginine = f (urea, NO, polyamines), the blocking created by the drug — analog of putrescine the synthesis of TNF- α and stop two processes:

- Development of the inflammatory process;
- Elimination of "side" effects on the cardiovascular system (hypotension/hypertension).

Finished dosage form is dissolved in ultra-low concentrations in physiological solution of the active substance analogue putrescine.

WWMA AG work is based on the fact that the process of septic lesions of the body first affect the organ system, the chief of which are the kidneys and vascular endothelium, and death may occur as a result of violations of the functional activity of the kidneys and vascular endothelium, which is the pathogenesis of low pressure and leads to death due to sudden decrease.

Synthesized by WWMA AG, the drug is an analogue of one of the polyamines (putrescine) and created as a result of deep study of the relevant metabolic processes at the cellular, intercellular and functional levels with the use of genetic engineering.

The working name of the drug – XR Coronavirus+.

Profile of preclinical studies to confirm the safety and efficacy of the drug XR Coronavirus+:

1. Acute toxicity
2. Subacute toxicity
3. Subchronic toxicity
4. Pharmacokinetics
5. Tolerance
6. Phagocytic
7. Immune response
8. Humoral immunity

CLINICAL TRIALS - PHASE 1

The drug to stimulate a specific immune response in complex infectious lesions — preparation “XR Coronavirus+” for the treatment of infectious diseases of complex etiology

SHORT EXTRACTION

The medication “XR Coronavirus+” dose-dependently stimulates in a wide dose range (3-25 ml/kg or 6-50 mg/kg once) the primary immune response, which is manifested in the increase of the amount of antibody forming cells (AFC) in the spleen and hemagglutinin in the blood of mice, mostly towards T- dependent than B-dependent antigen. Maximum stimulation of the immune response is achieved when the medication at a dose of 12.5 ml/kg (25 mg/kg) is injected intramuscularly. However, even with the 2-fold dose increase (0.5 ml/mouse) there is no change to immunosuppressive effect, which significantly differentiates it from the majority of currently known immune-modulators.

Immune-modulatory effect of the study medication is not significantly dependent on the sex of the animals. It is equally expressed in both high- and low-responsive mice to T-dependent antigen making it a promising medication for testing as a phenotypic corrector of the immune response, i.e., to correct genetically controlled defects in the immune system. The highest immune-modulatory effect of the medication is observed during the inductive phase of immunogenesis with optimal antigenic load.

Immune-modulatory potential of the medication allows its clinical usage by a special scheme 2x1 times/per week plus multiple times a day, plus a repeat during 5 weeks. At the same time the strain of the immune system and induction of non-specific lympho-proliferation in the spleen are not predicted.

In experimental models of immune deficiency in animals caused by cyclophosphamide, an immune modulator in doses of 1.25-5 ml/kg (2.5-10 mg/kg) cancels immunosuppression induced by a cytostatic agent and restores the content of AFC in the spleen to the level above the background parameters. The medication completely corrects immunosuppression in CBA mice caused by the external ionizing radiation.

STUDY PATIENTS

Twenty patients (as a model profile for testing the drug were selected — as the most difficult pathologies — chronic hiv/aids and acute-sepsis) participated in the clinical trials. According to the guidelines for the conduct of clinical trials 6 people voluntarily ceased their participation in the trials.

By age and sex (men comprised 60% of all participants in clinical trials), the group of examined patients was representative of the average of the city N. It included adult patients aged 20-39. Men accounted for 60% of the total number of patients (Table 1).

TABLE 1: CHARACTERISTICS OF HIV INFECTED PARTICIPANTS

Characteristics of HIV infected patients, who participated in the clinical trials of “XR Coronavirus+” according to age, height and body weight

Age, years	Number of examined patients	Height, cm	Number of examined patients	Weight, kg	Number of examined patients
15-19	0	Less than 149	1 (5%)	Less than 50	2 (10%)
20-24	4 (20%)	150-159	5 (25%)	50-59	9 (45%)
25-29	13 (65%)	160-169	6 (30%)	60-69	3 (15%)
30-34	1 (5%)	170-179	5 (25%)	70-79	5 (25%)
35-39	2(10%)	180-189	3 (15%)	80-89	1 (5%)

TABLE 2: CLASSIFICATION OF PARTICIPANTS BY STAGE OF HIV INFECTION

Table outlines classification of the examined patients, who participated in the clinical trials of “XR Coronavirus+”. Clinical trial participants were divided into groups depending on the stage of HIV infection.

№	Form of HIV infection	Number of examined patients
1	Latent	9
2	generalized lymphadenopathy	4
3	pre-AIDS	4
4	AIDS	3
	Total	20

The time-line of infection in the study group of patients ranged from 3 to 7 years.

In 3 patients sexual transmission was assumed; 17 patients were affected by injections (drug addiction). From the patients' testimony, the duration of drug abuse in 14 examined patients was more than 4 years. 16 patients are regularly checked at a dispensary. Current drug abuse is denied by 12 patients, while the rest take drugs occasionally. All patients were warned about the necessity to stop the drug abuse during the conduct of clinical trials of the medication.

TABLE 3: CLASSIFICATION OF PARTICIPANTS BY VIRAL LOAD

HIV infected clinical trial "XR Coronavirus+" participants were classified into groups depending on the viral load and clinical and laboratory parameters.

Groups	Group Characteristic	Number of HIV Infected Patients
1	Low viral load - LVL (less than 10000 RNA copies/ml)	12
2	Medium viral load - MVL (1000-100000 RNA copies/ml)	3
3	High viral load - HVL (>100000 RNA copies/ml)	5

Among the clinical symptoms of HIV infection the following were observed: dry cough (3 people), long low-grade fever (3 people), weight loss (6 people), difficult healing wounds (2 people), different disorders such as nausea, diarrhea, loss of appetite (10 people). No clinical evidence for the presence of malignant tumors was found.

Clinical examination of all patients participating in the trials of "XR Coronavirus+" did not reveal any essential organ pathology: the condition of patients was assessed as satisfactory.

All HIV infected patients were on an outpatient basis and under dispensary observation. In the course of the examination and before trials 3 patients were excluded from the trial participation: in one case because of the malignancy; the other two were excluded due to the prolonged subfebrile state of unclear origin. In AIDS patients at the beginning of the trials no active opportunistic diseases were observed.

CLINICAL LABORATORY EVALUATION OF TOLERABILITY, SAFETY AND EFFICACY OF “XR CORONAVIRUS+” DYNAMICS OF PERIPHERAL BLOOD PARAMETERS

As the data show the number of red blood cells and hemoglobin in the blood of patients was not reliably changed as a result of the application of “XR Coronavirus+”. No significant differences were observed in the groups depending on the virus concentration.

Only the color index reliably ($p < 0.05$) increased by the third injection in the whole group, as well as in patients with a virus concentration less than 10000 RNA copies/ml.

By the third injection the amount of platelets in the whole group decreased ($p < 0.05$) but it remained within the species norm.

The amount of leukocytes, eosinophils, rod-nuclear and segmented leukocytes, monocytes, and lymphocytes during clinical trials remained practically unchanged.

A reduced number of lymphocytes in patients with the load exceeding 100000 copies of RNA/ml compared to the total group and a tendency towards an increase in their number by the third injection should be noted.

During the application of “XR Coronavirus+” ESR fluctuations were revealed, and toward the end of clinical trials there was a reliable decrease in ESR in the whole group of HIV infected patients (19.5 ± 9.4 before the application of “XR Coronavirus+” and 12.9 ± 7.8 after the last injection, $p < 0.01$).

Thus, in the whole group, regardless of the virus concentration in the blood of HIV infected patients when “XR Coronavirus+” was applied no significant deviations in peripheral blood parameters were observed.

The decrease in the amount of platelets and lymphocytes at the viral load exceeding 100000 RNA copies/ml was reliable but it was within the norm indicating the absence of toxicity and side effects of this medication.

Dynamics of biochemical parameters of blood serum:

Main parameters of the biochemical status of the 20 clinical trial participants were studied.

It was revealed that in the whole group, and irrespective of the viral load, the amount of total protein in the serum of patients did not differ reliably. However, by the second injection and one month after it, before the third injection, the amount of total protein reliably ($p < 0.05$) decreased compared to the initial value almost regardless of the viral load.

Urea level after the first injection of the medication increased and remained the same for one month by the third injection. This is observed both in the general group and at low and medium viral loads. It should be noted that despite the reliable increase the concentration of urea did not exceed the clinical norm.

The same dynamics were true for blood glucose. One month after the medication was injected a reliable increase in glucose was observed. Seven days after the second injection the concentration of glucose in serum decreased but by the third injection, it reliably increased again to the normal upper border.

The concentration of cholesterol, on the contrary, by the third injection was reliably lower than before the injection of “XR Coronavirus+” in HIV infected patients with low and medium viral load.

Total bilirubin concentration in the examined patients did not change during the treatment and was not associated with the viral load parameters.

As for the level of aminotransferases, the activity of ALT and AST in the whole group reliably increased seven days after the second injection and remained the same until the third injection. Similar dynamics were observed at the viral load less than 10000 RNA copies/ml.

There is a tendency ($p>0.05$) to an increase in aminotransferase levels in patients with the virus load exceeding 10000 RNA copies/ml.

Thus, the application of “XR Coronavirus+” is characterised by variations of biochemical parameters in serum: decrease in total protein, cholesterol, increase of glucose and aminotransferase levels ($p<0.05$).

However, it should be noted that all parameters of the biochemical status did not exceed the conventional norms indicating the absence of hepato- and nephrotoxicity of the study medication.

**IMMUNOLOGICAL EVALUATION OF THE EFFICACY OF "XR CORONAVIRUS+"
IMMUNE STATUS AND THE EVALUATION OF THE PATIENTS DURING THE
APPLICATION OF "XR CORONAVIRUS+"**

The group of examined patients did not differ from the control group by quantitative content of leukocytes.

In HIV infected patients there was a reliable increase in relative and absolute lymphocytes: $42.31 \pm 3.34\%$ and $2.30 \pm 0.24 \cdot 10^9/l$ vs. $32.8 \pm 1.29\%$ and $1.65 \pm 0.11 \cdot 10^9/l$, respectively.

Immunological phenotype (Table 4) of peripheral blood mononuclear cells in HIV infected patients had the changes typical for HIV infection: a decrease in relative and absolute amount of CD3+- and CD4+- lymphocytes, increase in absolute and relative amount of CD8+- lymphocytes.

The immunoregulatory index value for the whole group was 7 ± 0.08 (in the control group 1.56 ± 0.11).

The content of B-lymphocytes (CD19+) was characterized by different changes compared to the control group but in the whole group there was a decrease in the number of B-lymphocytes.

TABLE 4: THE PHENOTYPE OF PERIPHERAL BLOOD MONONUCLEAR CELLS OF PATIENTS WITH HIV INFECTION

Parameter	Control group (n = 25)	HIV infected (n = 20)
Leukocytes, $\times 10^9/l$	5.0 $\pm$ 0.24	5.41 $\pm$ 0.41
Lymphocytes, %	32.8 $\pm$ 1.29	42.31 $\pm$ 3.34*
Lymphocytes, $\times 10^9/l$	1.65 $\pm$ 0.11	2.30 $\pm$ 0.24*
CD3, %	70.13 $\pm$ 1.21	59.28 $\pm$ 2.65*
CD3, $\times 10^9/l$	1.5 $\pm$ 0.11	1.3 $\pm$ 0.15
CD4, %	42.25 $\pm$ 1.53	27.18 $\pm$ 3.06*
CD4, $\times 10^7/l$	0.9 $\pm$ 0.08	0.70 $\pm$ 0.12
CD8, %	29.04 $\pm$ 1.45	41.24 $\pm$ 2.74*
CD8, $\times 10^9/l$	0.64 $\pm$ 0.05	1.04 $\pm$ 0.16*
CD4/CD8	1.56 $\pm$ 0.11	0.7 $\pm$ 0.08*
CD19, %	12.61 $\pm$ 0.94	6.11 $\pm$ 0.59*
CD19, $\times 10^9/l$	0.3 $\pm$ 0.03	0.2 $\pm$ 0.03*

- *differences are reliable compared to the control group, $p < 0,001$.

The analysis of the immunological phenotype of peripheral blood mononuclears in HIV infected patients depending on the viral load is particularly interesting (Table 5).

The amount of white blood cells, relative and absolute lymphocyte counts steadily decline when the viral load grows. The differences between mean values for the groups with low (LVL) and high (HVL) viral loads are reliable, $p < 0,05$ ($5.82 \pm 0.56 \times 10^9/l$, $46.70 \pm 4.84\%$ and $2.30 \pm 0.24 \times 10^9/l$ vs. $3.71 \pm 0.1 \times 10^9/l$ (?/n, $27.0 \pm 1.5\%$ and $1.0 \pm 0.01 \times 10^9/l$, respectively).

Similar tendencies were observed for the immune-regulatory subpopulations of T-lymphocytes.

TABLE 5: THE PHENOTYPE OF PERIPHERAL BLOOD MONONUCLEAR CELLS OF PATIENTS WITH HIV INFECTION DEPENDING ON THE VIRAL LOAD

Parameter	HIV Infected	Viral Load		
		<10 ⁴	10 ⁴ -10 ⁵	10 ⁴ -10 ⁵
Leukocytes, x10 ⁹ /l	5.41±0.41	5.82±0.56	5.63±1.69	3.71±0.11*
Lymphocytes, %	42.31±3.34	46.70±4.84	33.00±5.69	27.0±1.5 *
Lymphocytes, x10 ⁹ /l	2.30±0.24	2.67±0.35	1.72±0.30	1.0±0.01*
CD3,%	59.28±2.65	58.39±4.20	51.50±6.46	63.9 ±6.2
CD3, x10 ⁹ /l	1.3±0.15	1.50±0.21	0.96±0.32	0.71±0.09*
CD4,%	27.18±3.0b	33.79±5.63	27.10±6.00	20.6±4.3
CD4, x10 ⁹ /l	0.70±0.12	0.94±0.19	0.56±0.10	0.3±0.1*
CD8,%	41.24±2.74	45.47±3.95	34.15±6.40	41.1±6.0
CD8, x10 ⁹ /l	1.04±0.16	1.28±0.25	0.67±0.26	0.35±0.12*
CD4/CD8	0.7±0.08	0.78±0.14	0.85±0.22	0.6±0.2
CD19,%	6.11±0.59	6.05±0.65	5.70±0.93	5.7 ±1.1
CD19, x10 ⁹ /l	0.2±0.03	0.17±0.37	0.11±0.04	0.07±0.02

*- differences are reliable compared to the control group, p< 0,001.

PBMC DYNAMICS OF PATIENTS WITH HIV INFECTION DURING THE APPLICATION OF "XR CORONAVIRUS+"

The monitoring results of the immunological phenotype of main populations of peripheral blood lymphocytes are presented in Table 6. The content of immune cells, in general, was at the same level.

The main changes during the observation period are related to the subpopulation content of peripheral blood lymphocytes.

In 12 patients with HIV infection there was a significant increase in index of CD4/CD8 mainly due to decrease in the absolute content of CD8+ lymphocytes.

This tendency is favorable and it may reflect a decrease in functional activity of T-cytotoxic lymphocytes contributing to the killing and development of subsequent apoptosis, not only of virus-infected CP4+ lymphocytes but intact cells as well.

The intensity of changes in the ratio of CD4+ and CD8+ lymphocytes in 10 HIV infected patients had a phase character: 3 weeks after the first injection the parameter reduced compared to the initial value but 3 weeks after the second injection it increased dramatically.

In the whole group of HIV positive patients reliable ($p < 0.01$) increase in the immunoregulatory index compared to the initial values was registered.

TABLE 6: PHENOTYPE DYNAMICS OF PBMC IN PATIENTS WITH HIV INFECTION DURING THE APPLICATION OF "XR CORONAVIRUS+"

Parameter	Study 1	Study 2	Study 3	Study 4
Leukocytes *10 ⁹ /l	5.41±0.41	4.59±0.26	6.01±0.49	5.70±0.34
Lymphocytes, %	42.3±3.34	37.18±3.58	39.21±3.56	40.0±2.71
Lymphocytes *10 ⁹ /l	2.30±0.24	1.63±0.16*	2.40±0.30	2.28±0.20
CD3,%	59.28±2.6	55.20±2.79	65.88±2.46	65.88±1.86
CD3, *10 ⁹ /l	1.3±0.15	0.9±0.12*	1.7 ±0.23	1.53±0.16
CD4,%	27.18±3.1	23.94±1.54	25.6±1.88	25.98±1.9
CD4, x10 ⁹ /l	0.70±0.12	0.38±0.04*	0.62±0.09	0.55±0.06
CD8,%	41.2±2.74	43.10±2,30 '	30.34±2.5*	31.5±1.99*
CD8, x10 ⁹ /l	1.04±0.16	0.7±0.10	0.82±0.15	0.77±0.12
CD4/CD8	0.72±0.4	0.61±0.06	0.97±0.11	0.91±0.09
CD19,%	6.11±0.59	10.76±0.89*	7.54±0.83	6.70±0.60
CD19, x10 ⁹ /l	0.2±0.03	0.2±0.02	0.2 ±0.04	0.14±0.02

DYNAMICS OF PARAMETERS OF PERIPHERAL BLOOD MONONUCLEAR CELLS DEPENDING ON THE VIRAL LOAD

Main parameters of peripheral blood immune-competent cells in patients with HIV infection with the low viral load ($<10000/\text{ml}$) are shown in Tables 7-9.

In the group of HIV infected patients with low viral load (LVL) during the application of “XR Coronavirus+” by the end of clinical trials, there was a reliable increase in one indicator — the relative content of CD8+ lymphocytes. In the follow-up period variations in the amount of peripheral blood lymphocytes, changes in the relative and absolute count of T-helper cells were observed. It should be noted that in this group of examined patients a clear tendency to increase in immunoregulatory index was noted, the lack of reliability is related to the small sample size.

TABLE 7: PHENOTYPE DYNAMICS OF PERIPHERAL BLOOD MONONUCLEAR CELLS IN HIV INFECTED PATIENTS WITH LOW ($<10^4$) VIRAL LOAD

Parameter	HIV Infected (n=12)			
	Study 1	Study 2	Study 3	Study 4
Leukocytes $110^9/l$	5.82±0.57	4.90±0.29	5.76±0.59	5.64±0.31
Lymphocytes, %	46.70±4.84	37.10±5.33	40.30±4.91	43.82±4.57
Lymphocytes $*10^9/l$	2.67±0.35	1.76±0.25*	2.51±0.49	2.50±0.31
CD3, %	58.39±4.20	61.19±4.92	70.37±2.68»	66.75±2.93
CD3, $*10^9/l$	1.50±0.21	1.07±0.20	1.84±0.38	1.68±0.25
CD4, %	33.79±5.63	24.20±1.77	28.57±1.88	27.69±1.98
CD4, $*10^9/l$	0.94±0.19	0.42±0.06*	0.74±0.15	0.67±0.08
CD8, %	45.47±3.95	47.00±4.25	31.01±4.06*	32.60±3.31*
CD8, $*10^9/l$	1.28±0.25	0.84±0.18	0.85±0.25	0.88±0.20
CD4/CD8	0.78±0.15	0.59±0.09	1.12±0.17	0.96±0.14
CD 19, %	6.05±0.65	9.92±1.42*	7.67±1.31	6.25±0.89
CD19, $*10^9/l$	0.17±0.37	0.16±0.03	0.15±0.04	0.15±0.02

1 reliable differences compared to the initial values, $p<0.05$.

In HIV infected patients with the medium viral load (MVL) by the end of clinical trials of “XR Coronavirus+” (Table 8) the following was observed: the tendency to increase in the amount of peripheral blood lymphocytes, reliable increase in the absolute number of CD3⁺-lymphocytes and B- lymphocytes (CD19⁺). Low reliability of these changes is certainly related to the extremely small sample size.

TABLE 8: PHENOTYPE DYNAMICS OF PERIPHERAL BLOOD MONONUCLEAR CELLS OF HIV INFECTED PATIENTS WITH MEDIUM (10^4 - 10^5) VIRAL LOAD

Parameter	HIV infected (n=12)			
	Study 1	Study 2	Study 3	Study 4
Leukocytes $\times 10^9/l$	5.63 $\pm$ 1.69	5.40 $\pm$ 0.93	5.00 $\pm$ 0.29	6.73 $\pm$ 1.42
Lymphocytes, %	33.00 $\pm$ 5.69	39.0 $\pm$ 10.69	43.67 $\pm$ 5.78	40.50 $\pm$ 3.01
Lymphocytes $\times 10^9/l$	1.72 $\pm$ 0.30	1.94 $\pm$ 0.43	2.19 $\pm$ 0.33	2.63 $\pm$ 0.46
CD3,%	51.50 $\pm$ 6.46	49.77 $\pm$ 4.75	65.73 $\pm$ 6.50	62.17 $\pm$ 5.79
CD3, $\times 10^9/l$	0.96 $\pm$ 0.32	0.93 $\pm$ 0.12	1.39 $\pm$ 0.36	1.67 $\pm$ 0.38*
CD4,%	27.10 $\pm$ 6.00	20.67 $\pm$ 2.41	28.25 $\pm$ 4.93	19.65 $\pm$ 2.28
CD4, $\times 10^9/l$	0.56 $\pm$ 0.10	0.41 $\pm$ 0.10	0.50 $\pm$ 0.10	0.50 $\pm$ 0.07
CD8,%	34.15 $\pm$ 6.40	41.27 $\pm$ 4.57	30.65 $\pm$ 4.96	26.62 $\pm$ 3.86
CD8, $\times 10^9/l$	0.67 $\pm$ 0.26	0.77 $\pm$ 0.10	0.69 $\pm$ 0.25	0.75 $\pm$ 0.22
CD4/CD8	0.85 $\pm$ 0.22	0.52 $\pm$ 0.09	0.99 $\pm$ 0.19	0.83 $\pm$ 0.24
CD 19,%	5.70 $\pm$ 0.93	1.43 $\pm$ 2.35	6.68 $\pm$ 2.24	6.35 $\pm$ 1.52
CD19, $\times 10^9/l$	0.11 $\pm$ 0.04	0.24 $\pm$ 0.10	0.11 $\pm$ 0.04	0.16 $\pm$ 0.03

* reliable differences compared to the initial parameters, $p < 0.05$.

The immune status parameters in the HIV infected patients with the high viral load (HVL) during clinical trials of "PPU-15" had more pronounced variations compared to two previous groups (Table 9). After the 4th injection of the medication there was a significant increase in the relative and absolute number of lymphocytes, number of T-lymphocytes (CD3+), there was a clear tendency to an increase in the absolute content of CD4+-lymphocytes.

TABLE 9: PHENOTYPE DYNAMICS OF PERIPHERAL BLOOD MONONUCLEAR CELLS IN HIV PATIENTS WITH HIGH ($>10^5$) VIRAL LOAD

Parameter	HIV infected (n = 5)			
	Study 1	Study 2	Study 3	Study 4
Leukocytes $\times 10^9/l$	3.73 $\pm$ 0.13	3.92 $\pm$ 0.27	8.73 $\pm$ 1.95*	4.93 $\pm$ 0.87
Lymphocytes, %	27.00 $\pm$ 1.53	28.00 $\pm$ 2.04	40.67 $\pm$ 14.6	37.33 $\pm$ 3.18*
Lymphocytes $\times 10^9/l$	1.00 $\pm$ 0.02	1.09 $\pm$ 0.07*	3.10 $\pm$ 0.72*	1.79 $\pm$ 0.17*
CD3,%	63.91 $\pm$ 6.18	46.92 $\pm$ 4.61	55.80 $\pm$ 8.95	65.85 $\pm$ 1.55
CD3, $\times 10^9/l$	0.71 $\pm$ 0.09	0.55 $\pm$ 0.07	2.03 $\pm$ 0.51	1.28 $\pm$ 0.10*
CD4,%	20.58 $\pm$ 4.33	20.03 $\pm$ 2.29	17.96 $\pm$ 4.40	20.60 $\pm$ 3.60
CD4, $\times 10^9/l$	0.28 $\pm$ 0.07	0.25 $\pm$ 0.04	0.60 $\pm$ 0.07*	0.40 $\pm$ 0.05
CD8,%	41.05 $\pm$ 6.04	37.03 $\pm$ 3.31	29.40 $\pm$ 7.17	35.80 $\pm$ 8.20
CD8, $\times 10^9/l$	0.35 $\pm$ 0.12	0.38 $\pm$ 0.06	1.11 $\pm$ 0.33	0.69 $\pm$ 0.12
CD4/CD8	0.60 $\pm$ 0.16	0.57 $\pm$ 0.08	0.65 $\pm$ 0.12	0.58 $\pm$ 0.03
CD 19,%	5.70 $\pm$ 1.14	12.30 $\pm$ 2.07*	9.20 $\pm$ 1.70	9.15 $\pm$ 2.55
CD19, $\times 10^9/l$	0.07 $\pm$ 0.02	0.11 $\pm$ 0.02	0.34 $\pm$ 0.13	0.18 $\pm$ 0.04

* - reliable differences compared to the initial parameters, $p < 0.05$.

CHANGES IN THE LEVEL OF IMMUNOGLOBULINS IN THE SERUM OF HIV INFECTED PATIENTS DURING THE APPLICATION OF "XR CORONAVIRUS+"

In HIV infected patients during the initial evaluation the concentration of immunoglobulins G and A (Table 10) was not practically different from the control values ($p > 0.05$). Reliable differences were registered only for IgM ($p = 0.05$).

TABLE 10: THE CONCENTRATION OF IMMUNOGLOBULINS IN HIV INFECTED PATIENTS ACCORDING TO THE VIRAL LOAD

Groups	Concentration of immunoglobulins		
	IgG (g/l)	IgA (g/l)	IgM (g/l)
Control	12.95±0.7	1.41±0.13	1.65±0.13
HIV infected	13.27±0.97	1.81±0.19	2.05±0.2*
1 (LVL)	12.81±0.69	1.66±0.28	2.14±0.13*
2(MVL)	11.45±0.82	1.86±0.28	1.75±0.47
3(HVL)	1.74±2.06	1.84±0.55	2.03±0.23*

* - reliable differences compared to the control values ($p = 0.05$)

IgM level was comparable with the values of the control group only in HIV infected patients with the medium viral load.

During the application of "XR Coronavirus+" there were changes in the concentration of different classes of immunoglobulins in HIV infected patients.

By the end of clinical trials the content of immunoglobulins was characterized by the following differences (Table 11): increased concentration of IgG, IgA, and decreased concentration of IgM.

The changes almost equally occurred in HIV infected patients with medium, low, and high levels of viral load.

TABLE 11: THE CONCENTRATION OF IMMUNOGLOBULINS IN HIV INFECTED PATIENTS DEPENDING ON THE VIRUS CONCENTRATION ONE MONTH AFTER THE THIRD

Injection of "XR Coronavirus+" (n = 20)

Examined groups	Concentration of immunoglobulins		
	IgG (g/l)	IgA(g/l)	IgM
Control group	12.95±0.7	1.41±0.13	1.65 ±0.13
HIV infected	15.99±1.2*	2.0±0.24*	1.93±0.11*
1 (LVL)	17.45± 1.67*	2.77±0.43*	2.15±0.14
2 (MVL)	16.2±3.02*	1.98±0.24	1. 8±0.18
3(HVL)	16.2±3.6*	1.87±0.59	1.79±0.14*

*- reliability of the differences compared to the control values (p <0.05)

EVALUATION OF NONSPECIFIC RESISTANCE PARAMETERS IN HIV INFECTED PATIENTS DURING THE APPLICATION OF "XR CORONAVIRUS+"

The phagocytic activity of peripheral blood neutrophils in HIV infected patients in the whole group did not differ from the control values (Table 12).

TABLE 12: THE PHAGOCYTIC ACTIVITY OF POLYMORPHONUCLEAR LEUKOCYTES IN HIV INFECTED PATIENTS ACCORDING TO THE VIRAL LOAD AFTER THE APPLICATION OF "XR CORONAVIRUS+"

(n = 20)

Examined group	Parameters of the phagocytic activity of neutrophils	
	PI,%	PN
Control	48.7 ±1.29	4.68±0.2
HIV infected	46.62±3.24	4.33±0.25
1 (LVL)	49.18±4.92	4.39±0.39
2 (MVL)	36.75±3.9*	3.84±0.14*
3 (HVL)	44.5±7.57	3.68±0.38*

* - reliability of the differences compared to the initial parameters, p <0.05

However, in the study of phagocytosis index — phagocytic index (PI) and the number of absorbed objects (PN) depending on the viral load it was found that PI in patients with medium and high levels of viral load was reliably lower.

After the application of "XR Coronavirus+" there was a reliable increase in the phagocytic index and phagocytic number (Table 13) in HIV infected patients. This stimulation is most pronounced for the patients with the low and medium levels of viral load. However for the patients with the high viral load there is a clear tendency to an increase in phagocytic activity of neutrophils.

TABLE 13: THE PHAGOCYTIC ACTIVITY OF POLYMORPHONUCLEAR LEUKOCYTES IN HIV INFECTED PATIENTS ACCORDING TO THE VIRAL LOAD AFTER THE APPLICATION OF "XR CORONAVIRUS+"

(n = 20)

Examined group	Parameters of the phagocytic activity	
	PI,%	PN
Control	48.7±1.29	4.68±0.2
HIV infected	59.1±2.87*	5.87±0.26*
1 (LVL)	60.6±3.17*	5.95±0.14*
2(MVL)	50.0±9.49	6.13±0.79*
3(HVL)	51.5±12.5	4.55±0.25

* - reliability of the differences compared to the initial values, p <0.05

In the course of the 4-month studies in HIV infected patients with different levels of viral load, who received injections of "XR Coronavirus+", the medication was found to increase the content of peripheral blood lymphocytes mainly due to the T-lymphocyte population.

An increase in the value of the immunoregulatory index with underlying significant decrease in the number of T-suppressors/cytotoxic lymphocytes is observed that given the pathogenetic role of these cells can be considered definitely positive.

The medication contributes to the increase in the concentration of immunoglobulins G and A, and decrease in immunoglobulin M. The positive effect of the medication "XR Coronavirus+" on the phagocytic activity of neutrophils is indisputable.

In general there is significantly higher efficacy of the medication in terms of the individual selection of the dose and frequency of injections, which was described in the subsequent studies.

Changes in the viral load in HIV infected patients during the application of "XR Coronavirus+"

The most specific indicator reflecting the condition of HIV infected patients is the viral load level.

In the first phase of clinical trials it was established that in most cases, along with the viral load decrease, cholesterol level decreases as well (Figure 3-5). This fact deserves special attention in terms of the new data on the possible mechanisms of the immunodeficiency virus entry into the cell.

As it turned out in the course of the studies conducted by the scientists of the National Institute of Allergy and Infectious Diseases in the United States (NIAID), the medications lowering blood cholesterol can significantly slow the spread of the human immunodeficiency virus (HIV).

According to the obtained data, the virus needs cholesterol “platforms” in order to remain on the surface of the cells. Consequently, it can be assumed that cellular cholesterol decrease may significantly lower the rate of the virus spread.

CONCLUSIONS

1. “XR Coronavirus+” has no toxic effects when administered parenterally to humans: blood, urine, and biochemical parameters throughout the observation period were stable and did not exceed the normal range.
2. “XR Coronavirus+” is well tolerated; during the clinical trials 20 examined patients did not report any adverse symptoms.
3. “XR Coronavirus+” has no immunotoxic effects: in the course of its application in HIV infected patients no inhibition of the immune system was observed.
4. “XR Coronavirus+” helps to decrease the viral load, which is typical for 95% of the examined patients infected with HIV.
5. A preliminary assessment of the efficacy of “XR Coronavirus+” confirms that its effect is implemented through the immune system: during its application in HIV infected patients there is an increase of lymphocytes, mainly due to T-cells, an increase in the immunoregulatory index, increase in the concentration of immunoglobulins G and A, growth of phagocytic activity of neutrophils.
6. The application of “XR Coronavirus+” is accompanied by a decrease in cholesterol level, which pathogenetically confirms the efficacy of the medication and is consistent with the literature data.
7. The application of “XR Coronavirus+” improves the general condition of HIV infected patients, which is confirmed by laboratory parameters: an increase in the color index of blood, lowering of ESR, etc.

According to the data obtained during phase 1 of clinical trials the medication “XR Coronavirus+” can be recommended for studies in the 2nd phase of clinical trials.

CLINICAL TRIALS – PHASE 2

CONCLUSIONS

Below are the conclusions during extended clinical trial with 300 people with medication “XR Coronavirus+”

1. “XR Coronavirus+” is effective against a wide range of acute and chronic infectious and inflammatory diseases accompanied by the changes in the immune system reactivity, including acute respiratory infections, viral hepatitis, HIV infection, chlamydial infections, candidiasis, etc.
2. The efficacy of “XR Coronavirus+” is determined by the duration of the infectious-inflammatory process and the initial state of the immune system: in acute processes the effect of the treatment with “XR Coronavirus+” develops within 3-10 days, for chronic processes the therapy should continue for at least 6 months, for chronic viral hepatitis C, herpes infection caused by serotypes 2,3,4,5, and HIV infection the therapy should be for at least 1-2.5 years. Therapy duration should be increased in cases of combined infectious-inflammatory pathology.
3. The efficacy of “XR Coronavirus+” is determined by the characteristics of infectious agents, i.e. antigenicity, immunogenicity of pathogen structures, its biological properties (the ability to transform into a form of rest, the intracellular development cycle, phenotypic variability), as well as its localization in the body, which is explained by the structural features of the immune system in different biotopes of the body.
4. The action of “XR Coronavirus+” is implemented at the levels of antigen perception, presentation and recognition, i.e. at the level of antigen-specific immune response.
5. The implementation of immunotropic effects of “XR Coronavirus+” is based on the enhancing of the transport through the cytoplasmic cell membrane of secreted and membrane-bound molecules, including those of class I and II, molecules of the contact adhesion, receptors for cytokines and other biologically active substances, the molecules of cytokines and other biologically active molecules. “XR Coronavirus+” being a metabolite of numerous biochemical processes of cells optimizes immunoregulatory processes in a particular individual targeting them at the antigen elimination (causative agent microorganism), i.e. the development of immune response.
6. “XR Coronavirus+” does not modify the immune response and the range of synthesized and secreted cytokines. The efficacy of therapy with “XR Coronavirus+” depends on the initial prestimulation/preactivation of the organism and dominant cytokine profile, which is primarily determined by the chemical properties of the antigen, its dose, penetration into the body and the main affected biotope. Thus, the efficacy of “XR Coronavirus+” manifested by the elimination of the antigen/infectious agent from the body depends on the immune system’s ability to develop protective reactions or develop immunopathophysiologic reactions.
7. “XR Coronavirus+” activates antigen-specific forms of immune response and produces a significant effect on the regenerative processes in cells of different types including red blood cells, which is manifested by leveling of the effects of anemia, thrombocytopenia, monocytopenia and recovery of the immune system ability to develop lymphocytic (antigen-specific) immune response against infectious agents long persisting in the body and having immunosuppressive effect.

8. The therapy with “XR Coronavirus+” requires an individual approach to the patient. The medication has the ability to activate antigen-specific immune response against all infectious agents in the body. The activation sequence of the immune response against different microorganisms is not the same as it is determined by the process localization, i.e. the presence of antigen presenting cells, process stage, and the density of microbial population. Thus, without the preliminary assessment of the immune status and the differential diagnosis of the patient at the beginning of treatment of infectious and inflammatory diseases it is quite difficult to determine the sequence of immunotropic effects of “XR Coronavirus+”.

9. “XR Coronavirus+” facilitates the transformation of latent infection to active disease. This is accompanied by the appearance of bright clinical symptoms due to the development of the effector phase of the immune response and increase of the microbial load. From this point “XR Coronavirus+” can be used for the differential diagnosis of diseases in a short intense course. The activation of the microbial population can be implemented in several ways, particularly when cytokines are used by microorganisms as growth factors as well as the activation of the host cell genome during the immune response and the formation of non-reactivity of cells.

10. During the therapy with “XR Coronavirus+” 2 main types of effects are observed: 1) intermittent (remitting) and 2) lytic. These two types of response are associated with: 1) the duration of the disease, 2) the phenotypic variability of the pathogen microorganism, 3) the spare capacity of the body associated with the plastic exchange, age, sex, and characteristics of neuroendocrine regulation. Lytic effect is more common in the treatment of acute infectious diseases, remitting – for chronic diseases.

11. In the case of chronic infectious and inflammatory disease the therapy with “XR Coronavirus+” takes a considerable period of time (at least 12-18 months) for full recovery. Certain attitudes of the patient and an adequate assessment of the expected result of the therapy are required, considering the entire spectrum of the patient’s diseases, age, and lifestyle.

Short courses of therapy can provide good adaptogenic effect (“doping” effect); prolonged course causes the effector stage of the immune response, and therefore, the destruction of microbes and infected cells.

Taking into consideration the dual nature of the immune response the possibility of a severe course of therapy in patients with decompensated state should be considered as well. The risk of the therapy with “XR Coronavirus+” in such cases is determined not by the medication properties but by the disease characteristics of the patient, his immunological reactivity and adaptive-compensatory capacity. The rate between the medication dose and the patient’s characteristics is the following: the more severe or plural the disease, the more cautious should be the intensive courses of therapy with “XR Coronavirus+”.

12. “XR Coronavirus+” has a complex effect, which is primarily due to the unity of the homeostasis regulation of the nervous, immune, and endocrine systems. Hence, “XR Coronavirus+” minimizes asthenovegetative syndrome, increases capability, normalizes the emotional state of patients with chronic infectious-inflammatory processes. There is normalization in case of the initial violation of the menstrual cycle in women, spontaneous rejection of conventional doping (smoking, excessive consumption of coffee, tea) in many patients. Body weight normalization is also likely.

13. “XR Coronavirus+” does not have allergenic properties. However if the patient is allergic the medication might activate the process in the typical form. The therapy with “XR Coronavirus+” for the patients with allergic processes did not increase the frequency of attacks, on the contrary, there were positive results in 2 cases of infectious-allergic bronchial asthma due to its direct impact on the cause of

the disease. If the patient has autoimmune processes (autoimmune thyroiditis, diabetes, etc.) the application of "XR Coronavirus+" is not limited as well.

14. Side effects during the clinical study of the medication "XR Coronavirus+" were not observed. Soreness during intramuscular injections was observed, which was a problem in rare cases (1 failure for the entire period of the clinical trials of the medication). Signs of infiltration in the soft tissues of intramuscular injection sites were not found in any patient, even during intensive courses of therapy.

THEORY OF NEW GENERATION INFLUENZA VACCINE (NGIV) CREATION

The creators of anti-flu vaccine of new generation are V.Laskavyi, S.Ivanenko and S.Markushin (owner of the technology is WWMA AG Switzerland) have developed a new theory based on material and energy-informational influence of viruses: all enveloped viruses (lipid-content influenza virus) contain single "basic" element (strictly defined sequence of amino acid), which is a carrier of very important information for functional activity of all family of flu viruses regardless of the strain and the mutation of RNA.

Using data from science-based proposal, the authors also have found a way of delivery and fixation of this element in immune-competent organs of the body laboratory animals and humans.

For these purposes, the authors have used their own theory and development on the theme "4-component redox-system of bioorganisms" (see note 4), section "Metabolomics: evolution, research, management."

HYPOTHESIS

The foundation of the above theory was based on the following hypotheses:

- Protein lipid interaction structures of influenza virus and its nature;
- The basic element of protein structure of the flu virus (one of the peptides having the specific amino acid sequence) must be of the same type (structure, sequence) all enveloped (lipid-containing) viruses of influenza, regardless of the composition and mutations of RNA;
- Redox system of the organism according to the authors (who found full confirmation in practice) is composed of three main components, three metabolic groups (amino, AFK — peroxide, methyl) and sulfur component, which in varying degrees, participate in the stabilization of the protein upon lipid interaction (the factors stabilizing proteins) and, therefore, play an important role in the formation of the protection mechanism bio-organisms viral invasions.

Based on these hypotheses we developed a research profile, planned to find a "basic" element of a strict sequence of amino acids recorded in one of the peptides — see Reference 1.

REFERENCES

REFERENCE 1

The lipid-containing enveloped viruses are composed of three components — lipids, proteins and nucleic acids (NK). NK and protein form the nucleocapsid, which is attached to the lipid shell. The hypothesis of the authors is that all these viruses have the same peptide, which secures the lipids, because the all shells are formed by the interaction with the cell membrane. The mechanism of interaction is the same. This peptide can affect the virulence of the virus and its maturation. Protein is divided into sections: peptides and amino acids. Peptides are short sequences of amino acids. Amino acids are single molecules. At one end sits the amino group, the other carboxyl group. In the middle is attached a side chain. Individual characteristics of amino acids are determined this side group. Certain metabolites in a group of metabolomics redox systems, influence of the side chain that alters and/or blocks the action of a viral peptide. We understand the stabilizing properties of these metabolites on protein-protein and protein-lipid interactions.

A protein's stability will be determined by a myriad offactors, some of which are listed below. This isn't a comprehensive list, of course, but we believe it will still provide some insight into the different factors and what it is they do.

REFERENCE 2

- PH can change the charge state of some amino acids, and thus alter the fold of the protein
- Ionic strength, or the concentration of ions in the solution, can act to weaken charge-charge interactions between amino acids in a protein, changing its fold
- Solute concentrations act as molecular crowders. By virtue of their repulsion from protein surfaces, they create a crowding effect that pushes the protein toward more compact folds
- Temperature and pressure (at biologically reasonable values, in this case) are thermodynamic variables that affect a proteins fold through changes to the energy of non-covalent interactions in the protein's fold, as well as the effect of entropy in the system
- Just as important, a variety of interactions with molecularly small ligands, other proteins or biomacromolecules, or protein complexes can also stabilize a protein
- These include an entire class of proteins called [*chaperones*](#), that play an important role in the cell. These bind proteins and stabilize the native structure of the protein
- These also include osmolytes — small molecules that are able to induce a global stabilization of a cell's proteome.

Finally, binding by a protein will usually involve a change in the fold of the protein, as it shifts to accommodate its binding partner/ligand.

METHODS AND INSTRUMENTS USED FOR DEVELOPMENT OF NGIV

When creating this formulation a number of well known methods were used, as well as proprietary processes developed by our team:

- Modeling method (based on theoretical knowledge and their practical confirmation) processes, for example lipid containing corona virus transmissible gastroenteritis of pigs (patents of the authors on the virus strain HV-96, deposited in the VGNKI, and a vaccine against TGES);
- The method of site-specific mutagenesis;
- The attenuating method of a influenza strain a donor to create NGIV;
- Methods of genetic engineering technology for the synthesis of proteins;
- Methods cloning of gene fragments of plasmid virus;
- Method of liquid chromatography;
- Method of control selected elements on their farm activity in a special bioreactor;
- Method for the industrial production of vaccine based on genetic engineering, including methods of transformation by introducing into a bacterium (specific strain used. Esherihi-if) for protein production

The development of this vaccine was conducted on the basis of biotechnological processes and genetic engineering technologies.

THEORETICAL BASIS AND MATHEMATICAL MODEL FOR HUMAN BODY HOMEOSTASIS CONTROL ARE DEVELOPED

The model combines the approaches of modern medicine and biology, but mathematically formalized and formulated in the form of a system of differential equations. It was found that the state of health can be controlled by ultra-small periodic effects. This explains and theoretically justifies the effectiveness of Ultra High Dilution drugs.

Development of a mathematical model of homeostasis is completed with explanation of the control mechanism of the body Redox-system and four cycles, determining its resiliency:

- the Krebs cycle (mitochondria);
- pentose-phosphate cycle;
- the urea cycle (amino-group);
- cycle of one-carbon compounds (methyl group).

The model allows to manage and optimize homeostasis through regulation of the next processes:

- apoptosis;
- proliferations;
- phagocytosis;
- synthesis of catecholamines;
- energy mechanism of mitochondrial activity;
- epigenetics mechanism;
- to manage energy-informational "space" of the organism

The model is a coherent hierarchical system, having evanescent waves in view of its dynamics and allows to define pathology and “pathological” homeostasis.

Developed methodology and program management of Redox system (RS) include two methods:

- Method 1 – with use of dedicated key-important (arhaic) metabolites as active substances dissolved in saline in ultra-low doses;
- Method 2 – synthesized small molecules that allow to control RS “proton strikes”

These methods, via periodic programmed small effects, provide optimization of Redox-system, which leads to:

- Optimal functioning of each cycle;
- Coherence – overlapping of cycles in the same rhythm in due limits;

The described model allows to define pathology:

Pathology is the adaptive reaction of body Redox-system on parameter changes of coherence (demolition of a single rhythm/exceeding the limits of changes of the rhythm), expressed in non-recurrent (and/or convulsive) or resonant oscillations.

The model allows to more precisely determine the role and place of symptomatic medicine in current paradigm provision of temporary local assistance to a specific organ or system of the body with simultaneous systemic dyscrasia/imbalance of homeostasis of organism as a whole.

Five organs and associated systems of the body comprise the described system:

- Bones;
- Skeletal muscle;
- Adipose tissue;
- Heart;
- Brain

The role of other organs and systems:

- The gastrointestinal tract – the “wood” of the body energy systems;
- Lung – “bellows” of this system;
- Kidneys – dragange of ash;
- Liver – joint of energy with reproduction;
- Spleen – additional hematopoietic organ;
- Pancreas – main subsidiary organ, regulator of glucose streams and fatty acids, the mediator of insulin.

The role of phosphorus in the body is clarified:

- is the main element of the system of the energy homeostasis management;
- connects Redox system with energy (mitochondria);
- being involved at cellular level in:

- a) production of phospholipids – proliferation;
- b) energy cycle — ATP production, main important element in the system of switching of mitochondrial activity: disable mitochondria (apoptosis) under anaerobic conditions by methylglyoxal (linked to methyl group, role of archaic metabolites);
- at a body level, phosphorus, accumulated by bones (phosphates), manages energy, hence comes the role of bones (bone marrow) as hematopoietic and lymphogenous body (in evolution, “modern” bones emerged together with three-chambered heart)

COHERENT SYSTEMS

A new class of non-hierarchical coherent systems is proposed.

The systems are featured as elements, which are their cycles, while the consistency of functioning of cyclic processes is described by the wave function. The existence of cyclic processes is confirmed in biosphere and biological systems.

An example of a cyclic process is oxidation-reduction reactions.

In the simplest case, they have the form of a linear oscillatory circuit "donor-acceptor". This circuit consists of three elements: the donor, the acceptor and the element that is transferred from the donor to the acceptor.

Such elements in the general sense can be electrons and protons. A linear RLC circuit is one wave. One can adjust the frequency and amplitude of vibrations and the direction of wave propagation. It is regulated by external influences.

The first closed oscillatory circuit consists of 4 elements: the donor, the acceptor and the hermaphrodite. The hermaphrodite has donor-acceptor properties in equal ratio. The fourth element is what is transferred. This circuit depends on the properties of the hermaphrodite, on its equilibrium. Hermaphrodite is the object of control. The control assumes the presence of the fifth element.

In the biosphere, there are cycles of carbon, sulfur, nitrogen and oxygen. In the common system, the hermaphrodite is a carbon cycle.

The same cycles build structures in biological system. Operation of the system requires a constant flow of hydrogen.

Gradually, hydrogen, methyl group, acetate, glucose become such a source. At the output, hydrogen is accumulated in the produced product — methane, acetate, glucose, iric acid. In the process of evolution, the number of cyclic processes, associated with the root structure, increases. However, all cycles work in concert.

A mathematical model of the system of coherent cyclic processes as a set of individual cycles with the same coefficient is proposed. Under small influences through certain intervals of time, the wave function of the system is supported in a uniform continuous state. Variation of coefficients leads to coherence mismatch, which is expressed in emergence of chaos, which generates non-periodic or resonant wave structures. The system can be restored to its former uniform state, with small impacts at specified intervals.